

Lecture # 3

Wednesday
13 March 2024
9-10 AM
4-5 PM

PH102

After Mid Semester Exams

by

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Chapter-5

Magnetostatics

Section 5.4

Magnetic Vector Potential

Magnetic Vector Potential ($\mathbf{A}$)

In electrostatics we have

$$\nabla \times \vec{E} = 0$$

This permit us to introduce a scalar potential 'V'

$$\vec{E} = -\nabla V$$

In Magnetostatics

$$\nabla \cdot \vec{B} = 0$$

From the standard theorem of vectors which state that:

If the divergence of a vector field ($\vec{F}$) vanishes (everywhere), then $\vec{F}$ can be expressed as the curl of a vector potential.

This invites the introduction of a vector potential $\vec{A}$

$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{B} = \nabla \times \vec{A}$$

This won't affect the condition $\nabla \cdot \vec{B} = 0$, since the divergence of a curl is always zero

$$\nabla \cdot \nabla \times \vec{A} = 0$$

From the Ampere's law

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

$$\nabla \times \vec{B} = \nabla \times (\nabla \times \vec{A})$$

$$\nabla \times \vec{B} = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$$

$$\nabla \times \vec{B} = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

$$\nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

As we know, the electric potential had a built-in ambiguity: we can add to V any function whose gradient is zero (which is to say, any constant), without altering the physical quantity E .

$$\vec{E} = -\nabla(V + V_0)$$

$$\vec{E} = -\nabla V - \nabla V_0$$



0

$$\vec{E} = -\nabla V$$

Hence we can add any function to ‘ V ’ whose gradient is zero without altering the $\vec{E}$

Likewise, we can add to the magnetic potential any function whose curl vanishes (which is to say, the gradient of any scalar), with no effect on B . We can exploit this freedom to eliminate the divergence of $\vec{A}$

$$\nabla \cdot \vec{A} = 0$$

Proof: Suppose that our original potential, A_0 , is not divergence-less.

If we add the gradient of ' λ '

$$\vec{A} = \vec{A}_0 + \nabla\lambda$$

The new divergence is

$$\nabla \cdot \vec{A} = \nabla \cdot \vec{A}_0 + \nabla^2\lambda \longrightarrow 1$$

$$\nabla \cdot \vec{A} = 0$$

$$\nabla^2\lambda = -\nabla \cdot \vec{A}_0$$

Let's compare the above with to Poisson's equation for Electrostatics,

One way to
couple the
equations 1 & 2

$$\nabla^2 V = -\frac{\rho}{\epsilon_0} \longrightarrow 2$$

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{r} d\tau'$$

$$\rho \rightarrow 0 \text{ at } \infty$$

This equation is identical to Poisson's equation with $\nabla \cdot \vec{A}_0$ in place of ρ/ϵ_0 as the source.

$$\lambda = \frac{1}{4\pi} \int \frac{\nabla \cdot \vec{A}_0}{r} d\tau'$$

$$\nabla \cdot \vec{A}_0 \rightarrow 0 \text{ at } \infty$$

If $\nabla \cdot \vec{A}_0$ does not go to zero at infinity, we'll have to use other means to discover the appropriate λ , just as we get the electric potential by other means when the charge distribution extends to infinity.

Now

$$\nabla \times \vec{B} = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

$$\begin{aligned} \nabla \times \vec{B} &= \nabla \times (\nabla \times \vec{A}) \\ &= \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} \end{aligned}$$

$$\nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

$$\nabla \cdot \vec{A} = 0$$

$$\nabla^2 \vec{A} = -\mu_0 \vec{J}$$

This is again nothing but the Poisson's equation

We know the solution of Poisson's equation

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \frac{\overrightarrow{J(r')}}{r} d\tau'$$

$$\vec{J} \rightarrow 0 \text{ at } \infty$$

Similarly for line and surface currents,

Line current

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \frac{\vec{I}}{r} dl' = \frac{\mu_0 \vec{I}}{4\pi} \int \frac{1}{r} dl'$$

Surface current

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{r} da'$$

We know the solution of Poisson's equation

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \frac{\overrightarrow{J(r')}}{r} d\tau'$$

$$\vec{J} \rightarrow 0 \text{ at } \infty$$

Similarly for line and surface currents,

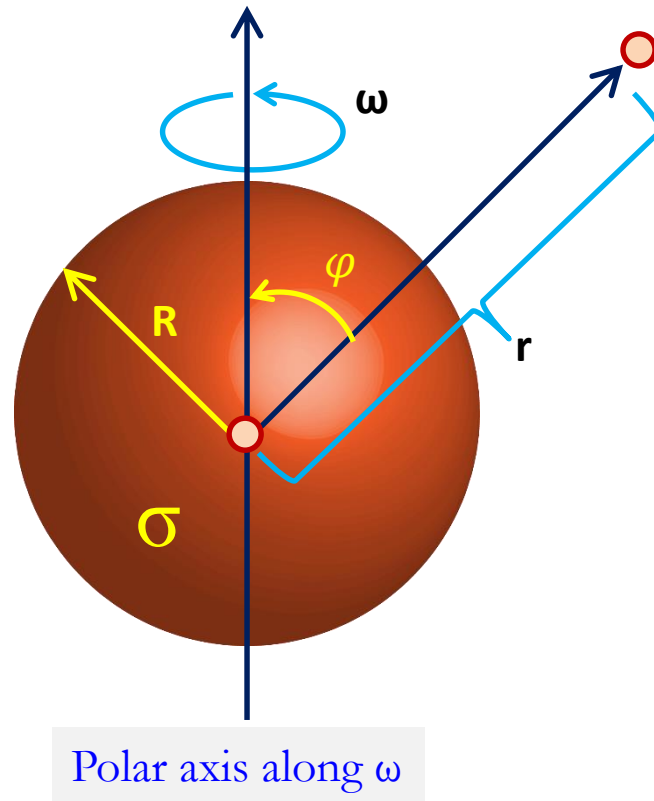
Line current

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \frac{\vec{I}}{r} dl' = \frac{\mu_0 \vec{I}}{4\pi} \int \frac{1}{r} dl'$$

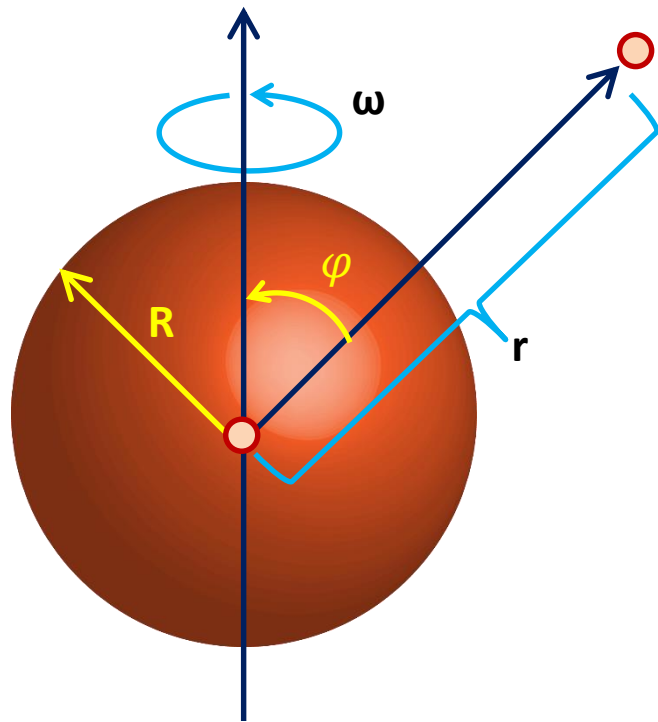
Surface current

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{r} da'$$

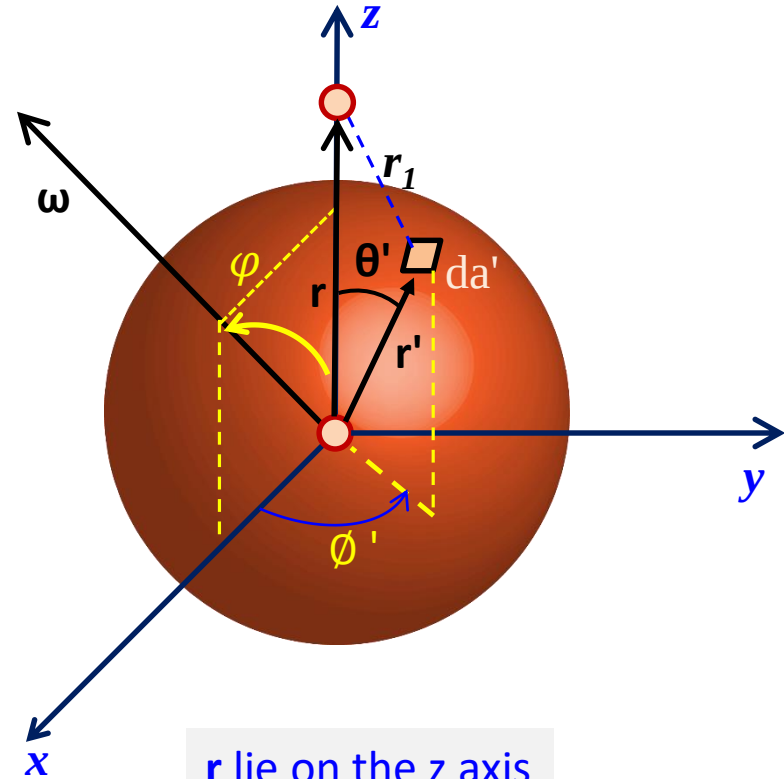
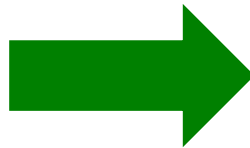
Example: A spherical shell, of radius R , carrying a uniform surface charge σ , is set spinning at angular velocity ω . Find the vector potential it produces at point $\mathbf{r}$.



Solution: It might seem natural to align the polar axis along ω , but in fact the integration is easier if we let $\mathbf{r}$ lie on the z axis, so that ω is tilted at an angle ψ . We may as well orient the x-axis so that ω lies in the xz-plane, as shown in the below figure (L.H.S)



Polar axis along ω

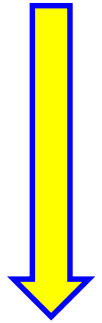


$\mathbf{r}$ lie on the z axis

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \frac{\overrightarrow{K(r')}}{r_1} da'$$

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int \frac{\overrightarrow{K(r')}}{r_1} da'$$

Here $K = \sigma V$



$$r_1 = \sqrt{R^2 + r^2 - 2 R r \cos \theta'}$$

$$da' = R^2 \sin \theta' d\theta' d\phi'$$

Velocity of a point r' in a rotating rigid body is given by $\omega \times r'$

$$\vec{v} = \vec{\omega} \times \vec{r}' = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \omega \sin \varphi & 0 & \omega \cos \varphi \\ R \sin \theta' \cos \phi' & R \sin \theta' \sin \phi' & R \cos \theta' \end{vmatrix}$$

$$\vec{v} = R\omega [-(\cos \varphi \sin \theta' \sin \phi')\hat{x} + (\cos \varphi \sin \theta' \cos \phi' - \sin \varphi \cos \theta')\hat{y} + (\sin \varphi \sin \theta' \sin \phi')\hat{z}]$$

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int_0^{\pi} \int_0^{2\pi} \frac{\sigma \overrightarrow{\omega} \times \overrightarrow{r'}}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \int_0^\pi \int_0^{2\pi} \frac{\sigma \overrightarrow{\omega} \times \overrightarrow{r'}}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} R^2 \sin \theta' d\theta' d\phi'$$

$$\overrightarrow{\omega} \times \overrightarrow{r'} = R\omega [-(\cos \varphi \sin \theta' \sin \phi')\hat{x} + (\cos \varphi \sin \theta' \cos \phi' - \sin \varphi \cos \theta')\hat{y} + (\sin \varphi \sin \theta' \sin \phi')\hat{z}]$$

↑
↑
↑

Since

$$\int_0^{2\pi} \sin \phi' d\phi' = \int_0^{2\pi} \cos \phi' d\phi' = 0$$

These terms contribute nothing and we are left with

$$\overrightarrow{A(r)} = -\frac{\mu_0 R^3 \sigma \omega \sin \varphi}{2} \int_0^\pi \frac{\cos \theta' \sin \theta'}{\sqrt{R^2 + r^2 - 2 R r \cos \theta'}} d\theta' \hat{y}$$

Substituting $u = \cos \theta'$, the integral becomes

Substituting $u = \cos \theta'$, the integral becomes

$$\begin{aligned} & \int_{-1}^{+1} \frac{u}{\sqrt{R^2 + r^2 - 2 R r u}} du \\ &= \frac{R^2 + r^2 + 2 R r u}{3 R^2 r^2} \sqrt{R^2 + r^2 - 2 R r u} \Big|_{-1}^{+1} \\ &= -\frac{1}{3 R^2 r^2} [(R^2 + r^2 + 2 R r)|R - r| - (R^2 + r^2 - 2 R r)(R + r)] \end{aligned}$$

If the point 'r' lies inside the sphere, then $R > r$, and this expression reduces to $(2r/3R^2)$; if 'r' lies outside the sphere, so that $R < r$, it reduces to $(2R/3r^2)$.

Noting that $(\vec{\omega} \times \vec{r}') = -\omega R \sin \varphi \hat{y}$ we have, finally.

$$\overrightarrow{A(r)} = \frac{\mu_0 R \sigma}{3} \overrightarrow{\omega} \times \vec{r}$$

For $r < R$

For points inside the sphere

$$\overrightarrow{A(r)} = \frac{\mu_0 R^4 \sigma}{3r^3} \overrightarrow{\omega} \times \vec{r}$$

For $r > R$

For points outside the sphere

$$\overrightarrow{\omega} \times \vec{r} = -\omega r \sin \theta \hat{\phi}$$

Taking ω along z-axis and the point r at (r, θ, φ)

Proof:

$$\overrightarrow{\omega} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & \omega \\ r \sin \theta \cos \varphi & r \sin \theta \sin \varphi & r \cos \theta \end{vmatrix}$$

$$= r\omega [-(\sin \theta \sin \varphi)\hat{x} + (\sin \theta \cos \varphi)\hat{y}]$$

$$\vec{\omega} \times \vec{r} = r\omega [-(\sin \theta \sin \phi)\hat{x} + (\sin \theta \cos \phi)\hat{y}]$$

$$= r\omega \sin \theta \underbrace{[-(\sin \phi)\hat{x} + (\cos \phi)\hat{y}]}_{\hat{\phi}}$$

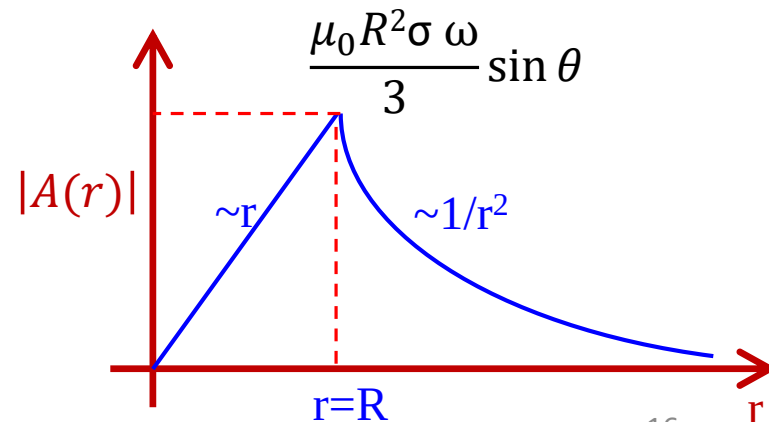
$$= r\omega \sin \theta \hat{\phi}$$

$$\vec{A}(r) = \frac{\mu_0 R \sigma \omega}{3} r \sin \theta \hat{\phi}$$

For $r < R$

$$\vec{A}(r) = \frac{\mu_0 R^4 \sigma \omega}{3r^2} \sin \theta \hat{\phi}$$

For $r > R$



Magnetic field inside the spherical shell is uniform

$$\vec{B} = \nabla \times \vec{A}$$

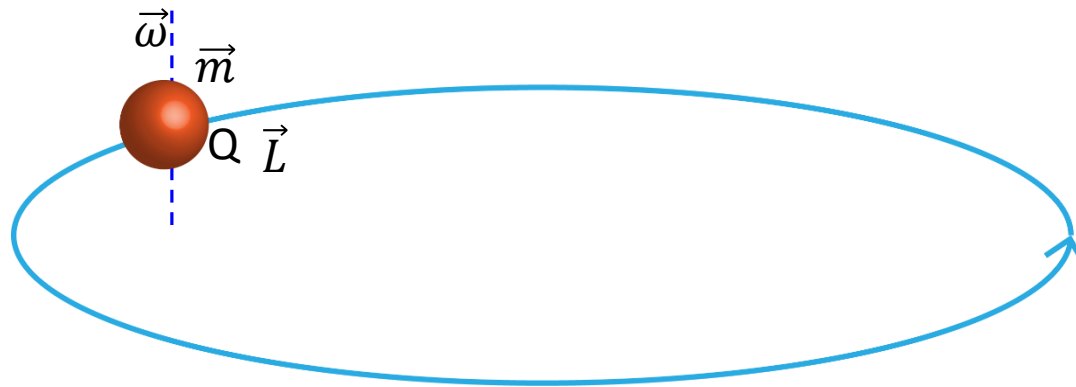
$$\vec{A}(r) = \frac{\mu_0 R \sigma \omega}{3} r \sin \theta \hat{\phi}$$

$$\vec{B} = \frac{2\mu_0 R \sigma \omega}{3} \underbrace{[\cos \theta \hat{r} - \sin \theta \hat{\theta}]}_{\hat{z}}$$

$$\vec{B} = \frac{2\mu_0 R \sigma \omega}{3} \hat{z}$$

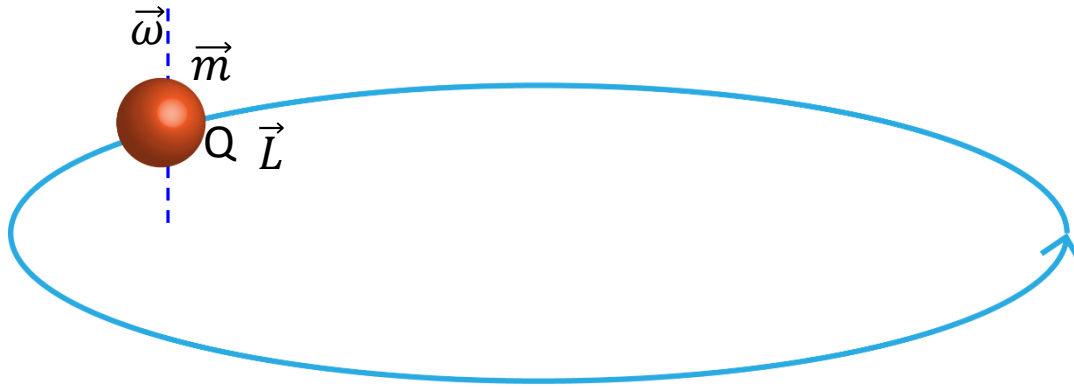
The Gyrometric Ratio

Let us consider a particle of charge Q and mass M describing a circle of radius a with angular velocity ω as shown in below figure. The equivalent current is



$$g = \frac{\text{magnetic moment}}{\text{Angular momentum}} = \frac{m}{L}$$

Let us consider a particle of charge Q and mass M describing a circle of radius a with angular velocity ω as shown in below figure. The equivalent current is



$$I = Q \times \text{frequency}$$

$$I = Q \times \frac{\omega}{2\pi}$$

This current generates a magnetic moment **m**

$$m = \frac{1}{2} Q \omega a^2$$

$$m = I \pi a^2$$

Let us take the angular momentum L of the particle into account.

$$L = M \omega a^2$$

The direction of L is given by the right hand rule.

$$m = \frac{1}{2} Q \omega a^2$$

For +ve charge

Depending upon the charge

$$m = -\frac{1}{2} Q \omega a^2$$

For -ve charge

For a positively charged particle the ratio

$$g = \frac{\text{magnetic moment}}{\text{Angular moment}} = \frac{m}{L}$$

$$g = \frac{\frac{1}{2} Q \omega a^2}{M \omega a^2}$$

$$g = \frac{Q}{2M}$$

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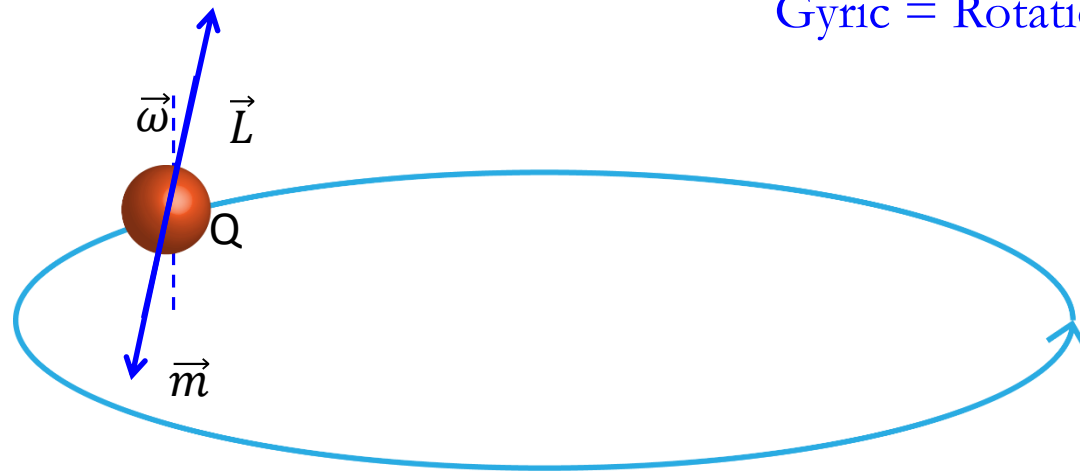
This ratio is called the Gyromagnetic ratio

Magnetogyric ratio

Similarly for a negative charged particle

$$g = -\frac{Q}{2M}$$

The negative sign indicates that the vectors $\mathbf{L}$ and $\mathbf{m}$ have opposite direction

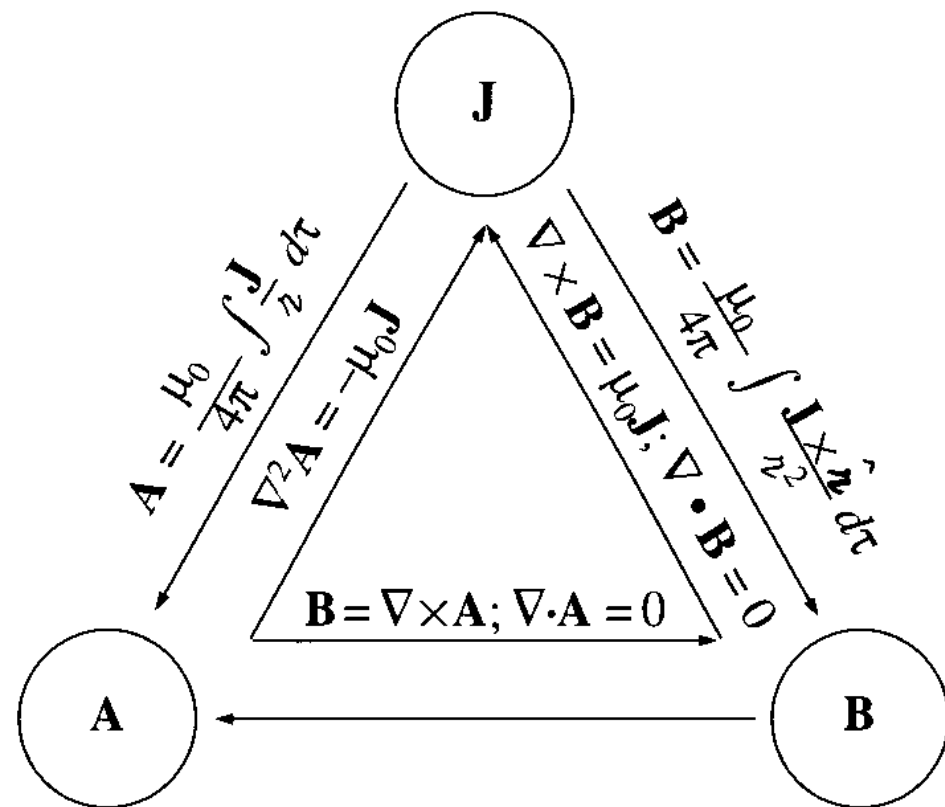


Gyric = Rotational (from Greek)

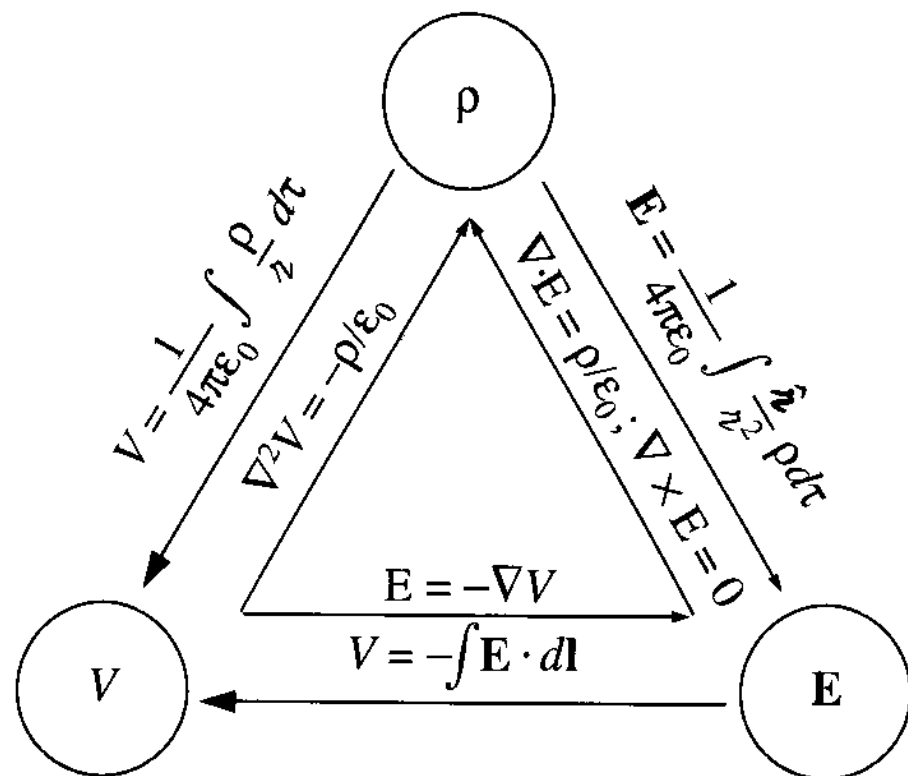
SI units radian per second per Tesla ($\text{rad} \cdot \text{s}^{-1} \cdot \text{T}^{-1}$) or, equivalently, the Coulomb per kilogram ($\text{C} \cdot \text{kg}^{-1}$).

Summary

Magnetostatics

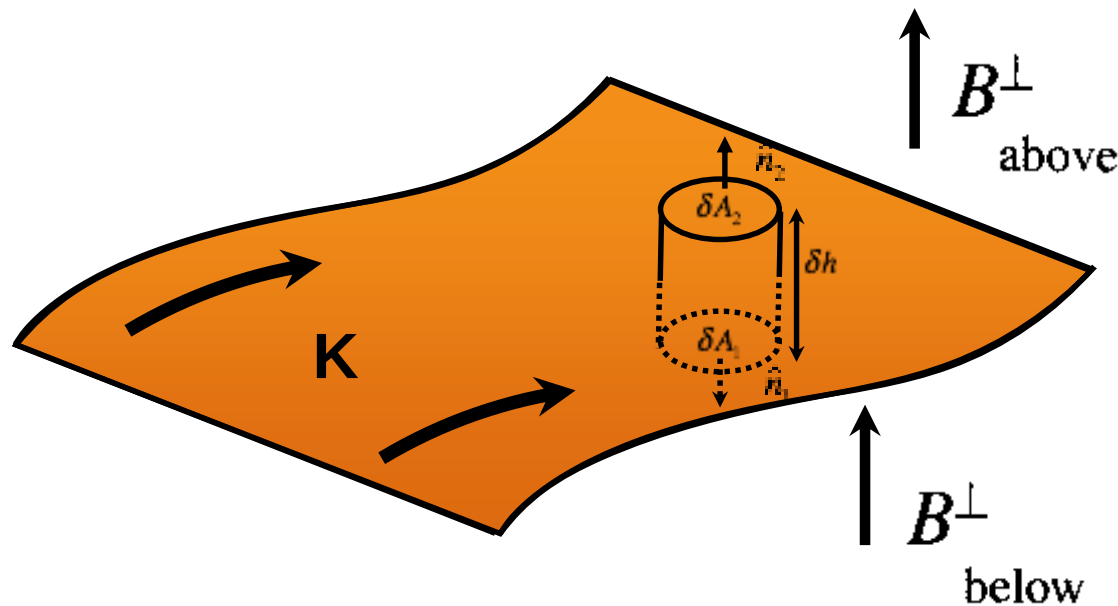


Electrostatics



Magnetostatic Boundary conditions

Just as the electric field suffers a discontinuity at a surface charge, so the magnetic field is discontinuous at a surface current.



$$\nabla \cdot \mathbf{B} = 0$$

$$\oint \mathbf{B} \cdot d\mathbf{a} = 0$$

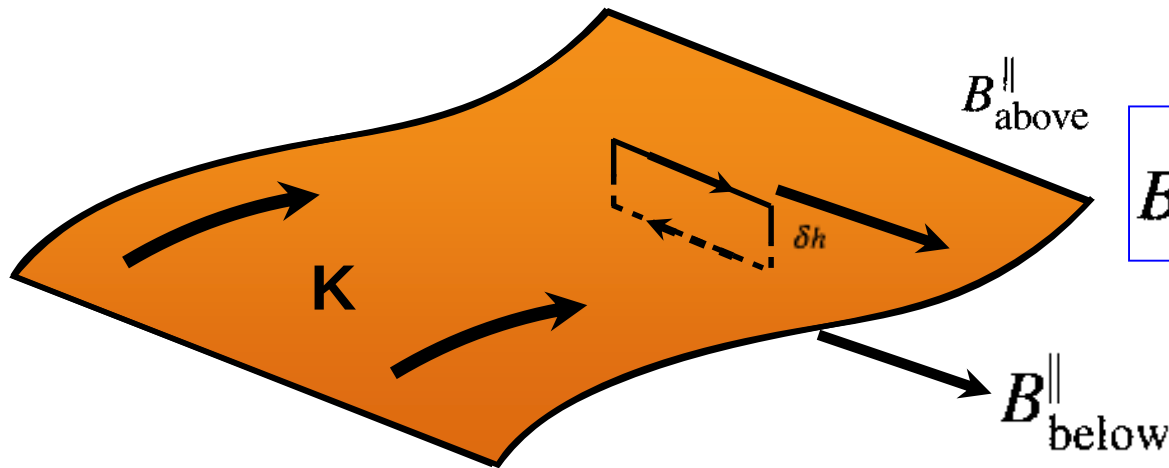
$$B_{\text{above}}^{\perp} - B_{\text{below}}^{\perp} = 0$$

Amperian loop running perpendicular to the current

$$\oint \mathbf{B} \cdot d\mathbf{l} = (B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel})l = \mu_0 I_{\text{enc}} = \mu_0 K l$$

$$B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} = \mu_0 K$$

$$\oint \mathbf{B} \cdot d\mathbf{l} = (B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel})l = \mu_0 I_{\text{enc}} = \mu_0 K l$$



$$B_{\text{above}}^{\parallel} - B_{\text{below}}^{\parallel} = \mu_0 K$$

In general



$$\mathbf{B}_{\text{above}} - \mathbf{B}_{\text{below}} = \mu_0 (\mathbf{K} \times \hat{\mathbf{n}})$$

$\hat{\mathbf{n}}$ is a unit vector perpendicular to the surface, pointing "upward."

Thus the component of $\mathbf{B}$ that is parallel to the surface but perpendicular to the current is discontinuous in the amount $\mu_0 K$. A similar Amperian loop running parallel to the current reveals that the parallel component is continuous. These results can be summarized in a single formula:

Boundary condition for $\mathbf{A}$

Like the scalar potential in electrostatics, the vector potential is continuous across any boundary:

$$\nabla \cdot \vec{A} = 0 \Rightarrow A_{above}^{\perp} = A_{below}^{\perp}$$

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_S \mathbf{v} \cdot d\mathbf{a}.$$

$$\nabla \times \vec{A} = \vec{B}$$

$$(\nabla \times \vec{A})_{above} - (\nabla \times \vec{A})_{below} = \mu_0(\vec{K} \times \hat{n})$$

$$(\hat{n} \frac{\partial}{\partial n} \times \vec{A})_{above} - (\hat{n} \frac{\partial}{\partial n} \times \vec{A})_{below} = \mu_0(\vec{K} \times \hat{n})$$

$$\frac{\partial \vec{A}_{above}}{\partial n} - \frac{\partial \vec{A}_{below}}{\partial n} = -\mu_0 \vec{K}$$

In electrostatics, if you are very far from a localized charge distribution it looks like a point charge and the potential is – to good approximation-
 $(1/4\epsilon_0\pi)Q/r$

Monopole
 $(V \sim 1/r)$

Dipole
 $(V \sim 1/r^2)$

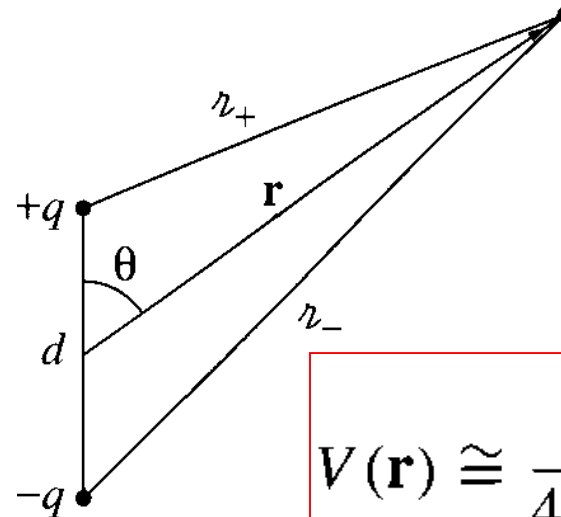
Quadrupole
 $(V \sim 1/r^3)$

Octopole
 $(V \sim 1/r^4)$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_+} - \frac{q}{r_-} \right)$$

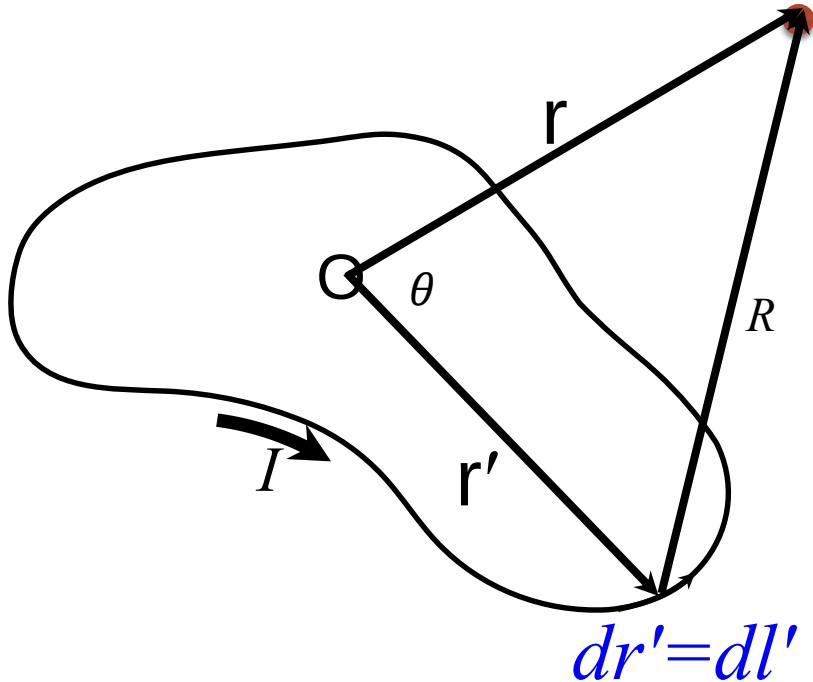
$$r^2 \left(1 \mp \frac{d}{r} \cos \theta + \frac{d^2}{4r^2} \right)$$

$$\frac{1}{r} \left(1 \mp \frac{d}{r} \cos \theta \right)^{-1/2} \cong \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right)$$



$$V(\mathbf{r}) \cong \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2}$$

Multipole Expansion of the Vector Potential



Vector potential of a current loop

$$\vec{A} = \frac{\mu_0 I}{4\pi} \oint \frac{1}{R} d\vec{l}'$$

$$R = (r^2 + r'^2 - 2rr' \cos \theta)^{1/2}$$

$$\frac{1}{R} = \frac{1}{(r^2 + r'^2 - 2rr' \cos \theta)^{1/2}}$$

$$\frac{1}{R} = \frac{1}{r \left(1 + \left(\frac{r'}{r} \right)^2 - 2 \frac{r'}{r} \cos \theta \right)^{1/2}}$$

For $r \gg r'$

$$\frac{1}{\vec{R}} \approx \frac{1}{r \left(1 - 2 \frac{r'}{r} \cos \theta \right)^{1/2}}$$

$$\frac{1}{\vec{R}} \approx r \left(1 - 2 \frac{r'}{r} \cos \theta \right)^{-1/2}$$

$$\frac{1}{\vec{R}} \approx r \left(1 + \frac{r'}{r} \cos \theta + \dots \right)$$


$$\vec{A} = \frac{\mu_0 \vec{I}}{4\pi} \oint \frac{1}{R} d\vec{l}'$$

$$\vec{A} = \frac{\mu_0 \vec{I}}{4\pi} \oint r \left(1 + \frac{r'}{r} \cos \theta + \dots \right) d\vec{l}'$$

$$\vec{A} = \frac{\mu_0 \vec{I}}{4\pi} \left[\underbrace{\frac{1}{r} \oint \vec{dl'}}_{\text{Monopole}} + \underbrace{\frac{1}{r^2} \oint (r' \cos \theta) \vec{dl'}}_{\text{Dipole}} + \dots \right]$$

Now the magnetic monopole term is always zero

$$\oint \vec{dl'} = 0$$

$$\vec{A}_{dipole} = \frac{\mu_0 \vec{I}}{4\pi r^2} \oint r' \cos \theta \vec{dl'}$$

$$\vec{A}_{dipole} = \frac{\mu_0 \vec{I}}{4\pi r^2} \oint (\hat{r} \cdot \vec{r'}) \vec{dl'}$$

$$\begin{aligned}\hat{r} \times \oint (\vec{r}' \times \overrightarrow{dr'}) &= \oint \hat{r} \times (\vec{r}' \times \overrightarrow{dr'}) \\ &= \oint [\vec{r}'(\hat{r} \cdot \overrightarrow{dr'}) - \overrightarrow{dr'}(\hat{r} \cdot \vec{r}')] \quad (\text{BAC-CAB})\end{aligned}$$

Now, $d[(\vec{r}'(\hat{r} \cdot \vec{r}'))] = \vec{r}'(\hat{r} \cdot \overrightarrow{dr'}) + \overrightarrow{dr'}(\hat{r} \cdot \vec{r}')$

$$\Rightarrow \vec{r}'(\hat{r} \cdot \overrightarrow{dr'}) = d[(\vec{r}'(\hat{r} \cdot \vec{r}'))] - \overrightarrow{dr'}(\hat{r} \cdot \vec{r}')$$

$$\begin{aligned}\hat{r} \times \oint (\vec{r}' \times \overrightarrow{dr'}) &= \oint [\vec{r}'(\hat{r} \cdot \overrightarrow{dr'}) - \overrightarrow{dr'}(\hat{r} \cdot \vec{r}')] \\ &= \oint d[(\vec{r}'(\hat{r} \cdot \vec{r}'))] - \oint (\hat{r} \cdot \vec{r}') \overrightarrow{dr'} - \oint \overrightarrow{dr'}(\hat{r} \cdot \vec{r}')$$

$$\oint d[(\vec{r}'(\hat{r} \cdot \vec{r}'))] = \oint d[(\vec{r}'(r' \cos \theta))] = 0 \quad \text{in case of closed path}$$

Therefore, $\hat{r} \times \oint (\vec{r}' \times \overrightarrow{dr'}) = -2 \oint (\hat{r} \cdot \vec{r}') \overrightarrow{dr'}$

$$\hat{r} \times \oint (\vec{r}' \times \overrightarrow{dr'}) = \oint \vec{r}' (\hat{r} \cdot \overrightarrow{dr'}) - \oint \overrightarrow{dr'} (\hat{r} \cdot \vec{r}')$$

$$\hat{r} \times \oint (\vec{r}' \times \overrightarrow{dr'}) = \oint d[(\vec{r}' (\hat{r} \cdot \vec{r}'))] - \oint (\hat{r} \cdot \vec{r}') \overrightarrow{dr'} - \oint \overrightarrow{dr'} (\hat{r} \cdot \vec{r}')$$

Zero

$$\hat{r} \times \oint (\vec{r}' \times \overrightarrow{dr'}) = -2 \oint (\hat{r} \cdot \vec{r}') \overrightarrow{dr'}$$

For present condition

$$\overrightarrow{dr'} = \overrightarrow{dl'}$$

$$\hat{r} \times \oint (\vec{r}' \times \overrightarrow{dl'}) = -2 \oint (\hat{r} \cdot \vec{r}') \overrightarrow{dl'}$$

$$\hat{r} \times \oint (\vec{r}' \times \overrightarrow{dl'}) = -2 \oint (\hat{r} \cdot \vec{r}') \overrightarrow{dl'}$$

$$\vec{A}_{dipole} = \frac{\mu_0 \vec{I}}{4\pi r^2} \oint (\hat{r} \cdot \vec{r}') \overrightarrow{dl'}$$

$$\vec{A}_{dipole} = -\frac{\mu_0 \vec{I}}{4\pi r^2} \left[\frac{1}{2} \left(\hat{r} \times \oint (\vec{r}' \times \overrightarrow{dl'}) \right) \right]$$

$$\vec{A}_{dipole} = \frac{\mu_0}{4\pi} \left[\frac{\vec{m} \times \hat{r}}{r^2} \right]$$

Here

$$\vec{m} = \frac{I}{2} \oint \vec{r}' \times \overrightarrow{dl'}$$

And

$$\vec{a} = \frac{1}{2} \oint \vec{r}' \times \overrightarrow{dl'}$$

$$\vec{m} = I \vec{a}$$

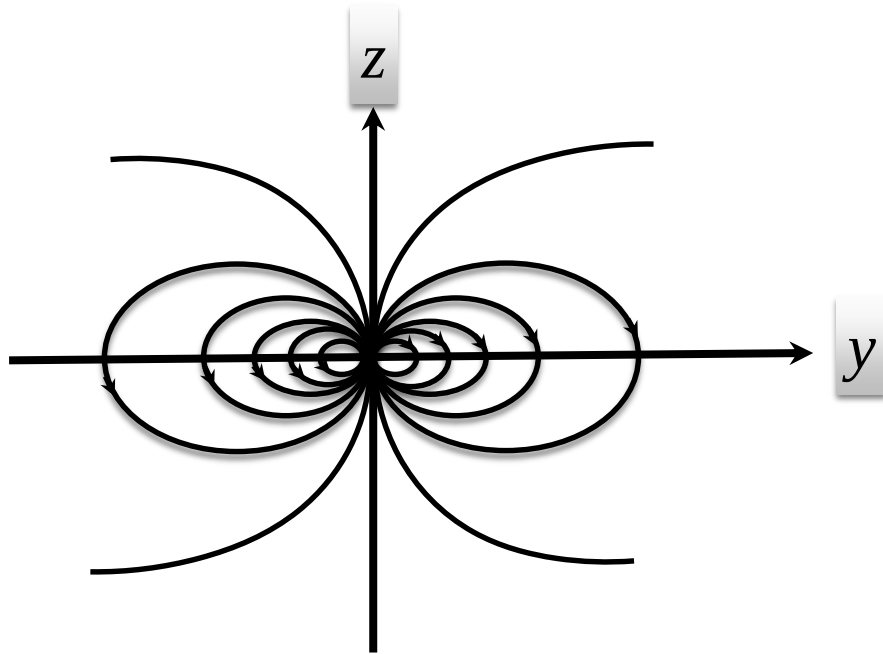
Vector Area, P-57,
Problem 1.61
Eq-1.107

Summarized Results

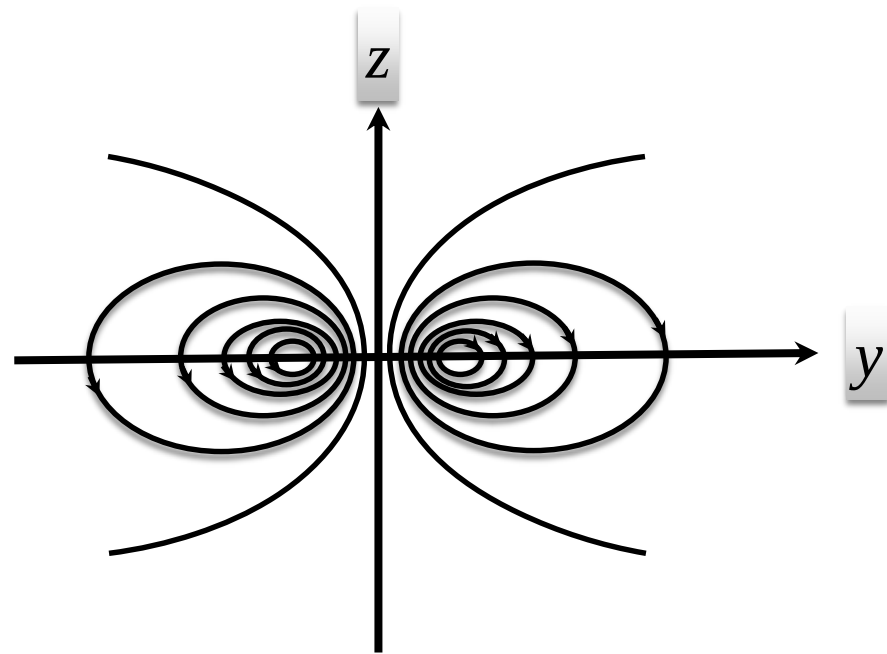
$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \left(\frac{m \sin \theta}{r^2} \right) \hat{\phi}$$

$$\vec{B} = \frac{\mu_0 m}{4\pi} \frac{1}{r^3} [2 \cos \theta \hat{r} + \sin \theta \hat{\theta}]$$

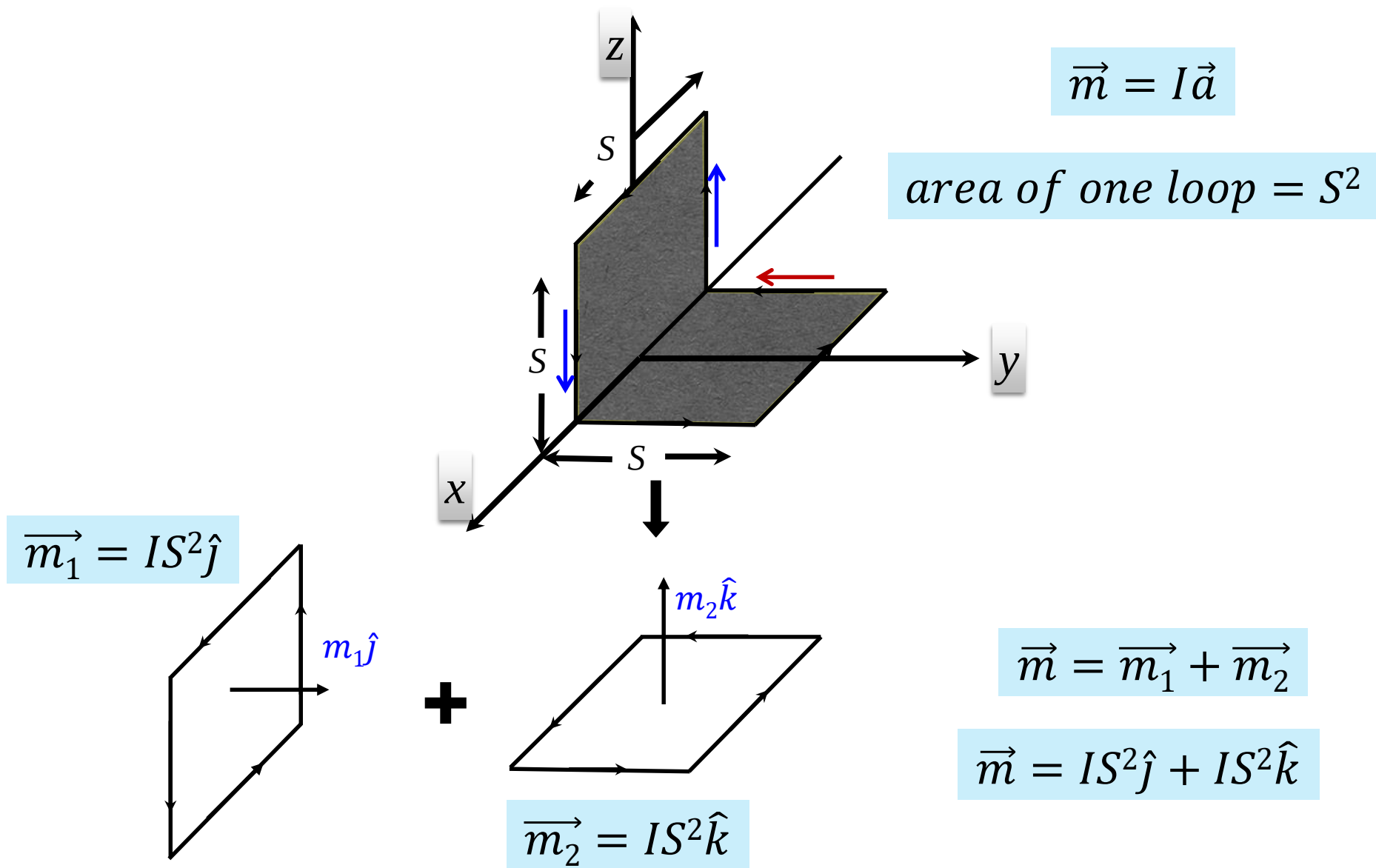
Field of a "pure" dipole



Field of a "physical" dipole



Example: Find the magnetic dipole moment of the "bookend-shaped" loop shown in Fig. All sides have length S , and it carries a current I .



Example: Calculate the magnetic flux crossing the surface $\varphi = \pi/2$, $1 \leq \rho \leq 2$, $0 \leq z \leq 5$ if the magnetic vector potential is given by $\vec{A} = -\frac{\rho^2}{4}\hat{z}$.

Solution:

$$\vec{B} = \nabla \times \vec{A}$$

For cylindrical co-ordinates (in general)

$$\nabla \times \vec{v} = \left(\frac{1}{\rho} \frac{dv_z}{d\phi} - \frac{dv_\phi}{dz} \right) \hat{\rho} + \left(\frac{dv_\rho}{dz} - \frac{dv_z}{d\rho} \right) \hat{\phi} + \frac{1}{\rho} \left(\frac{d}{d\rho} (\rho v_\phi) - \frac{dv_\rho}{d\phi} \right) \hat{z}$$



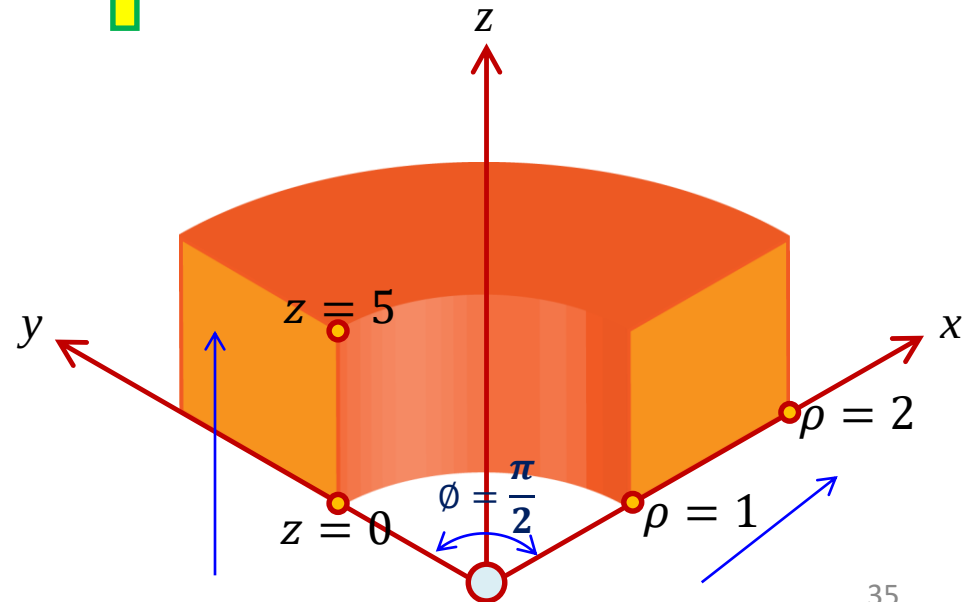
Here

$$\vec{A} = -\frac{\rho^2}{4}\hat{z}$$

$$\vec{B} = -\frac{dA_z}{d\rho}\hat{\phi}$$

$$\vec{B} = \frac{\rho}{2}\hat{\phi}$$

Since A has only
z-component and
only ρ
dependence

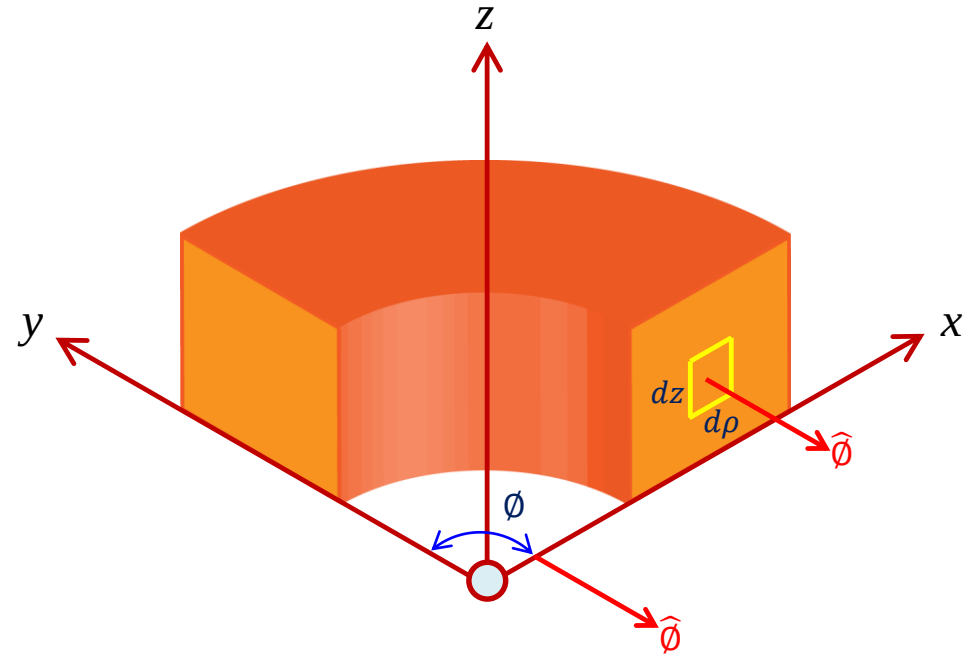


$$\overrightarrow{ds} = d\rho \, dz \, \hat{\phi}$$

$$\text{Magnetic flux} = \int \vec{B} \cdot \overrightarrow{ds}$$

$$\vec{B} = \frac{\rho}{2} \hat{\phi}$$

$$= \int_{z=0}^5 \int_{\rho=1}^2 \frac{1}{2} \rho \, d\rho \, dz$$



$$\Phi = \frac{15}{4} \text{ Webers}$$